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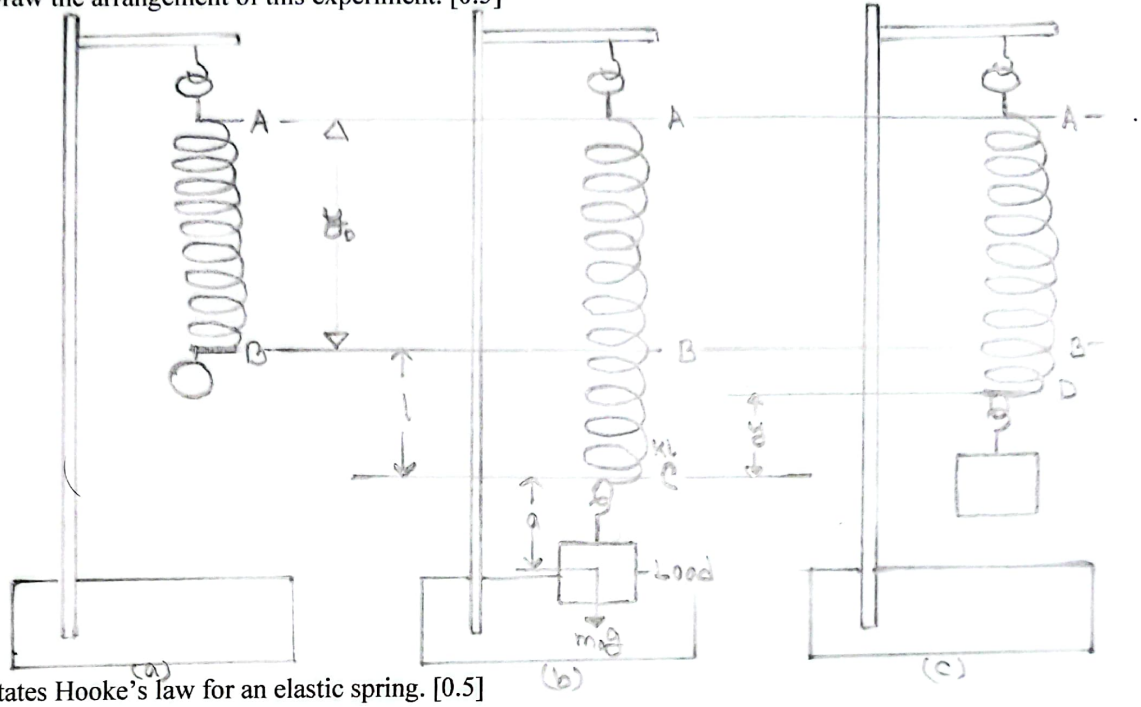
Experiment no: 4

Name of the Experiment: Determination of the spring constant and effective mass of a given spiral spring.

Questions on Theory

*1) Draw the arrangement of this experiment. [0.5]

Ans:



*2) States Hooke's law for an elastic spring. [0.5]

Ans: Hook's law state that for small deformation, stress is directly proportional to strain.

*3) Suppose, an external force is applied on a spring to stretch it. Extension of the spring is *l*. If the spring constant is *k* then what is the restoring force of the spring due to its elasticity? [0.5]

Ans: According to Hooke's law, within the elastic limit the magnitude of this restoring force is directly proportional to the extension of the spring and the direction of this restoring force is always towards the equilibrium position. If K is the spring constant of the spring and l is the extension of the spring, then restoring force = $-Kl$.

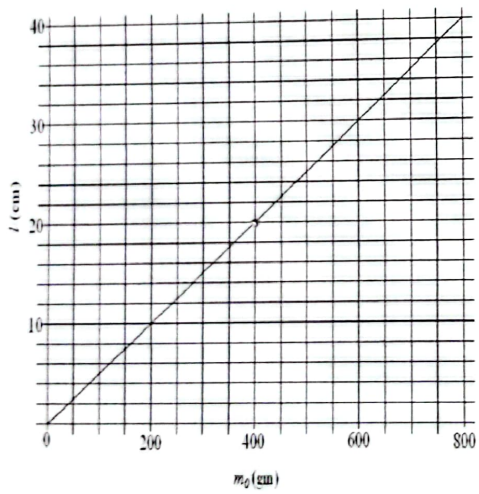
*4) If a load of m_0 is connected with the bottom end of a vertically suspended spring, the extension of the spring is l. The spring constant is k. At the equilibrium condition what is the relationship among m_0, k, l and g? [0.5]

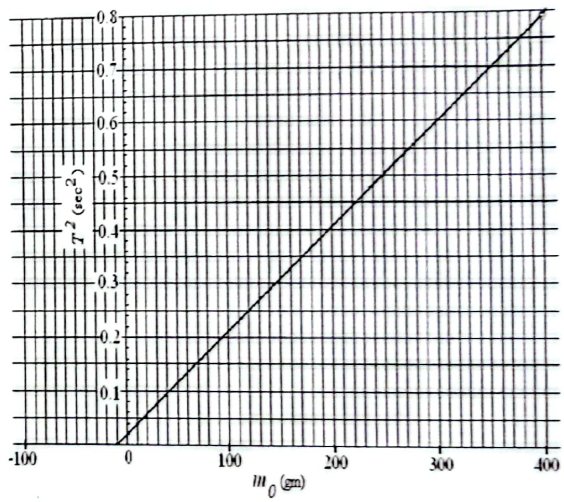
Ans: If a load of m_0 is connected with bottom end of a vertically suspended spring, the extension of the spring is l. The spring constant is K. At the equilibrium condition the relationship among m_0, K, l and g is, $m_0 g = Kl \Rightarrow l = \frac{g}{K} m_0$.

5) What is the effective mass of the spring? (See Appendix A/textbook/websites) [1]

Ans: The effective mass of the spring is one-third of its actual mass. Effective mass, $m' = \frac{m}{3}$. Effective mass is the mass added to the spring for correctly predicting system behaviour.

6) An l vs. m_0 and a T^2 vs. m_0 graphs are shown below. Work out the spring constant and effective mass. [2]





[You may use additional page(s) to answer this question]

Ans: From the graphs we get,

$m_0 = 400$ gm

$l = 20$ cm

$g = 9.8\ ms^{-2}$

$= 980\ cms^{-2}$

$\therefore m_0 g = Kl$

$\Rightarrow K = \frac{m_0 g}{l} = \frac{400 \times 980}{20} = 19600\ gms^{-2}$

From graphs we get,

$m_0 = 400\,gm$

$T^2 = 0.785\,s^2$

$K = 19600\,gm\,s^{-2}$

$$\therefore T = 2\pi\sqrt{\frac{m_0+m'}{K}}$$

$$\Rightarrow T^2 = 4\pi^2\frac{m_0+m'}{K}$$

$$\Rightarrow \frac{T^2K}{4\pi^2} = m_0+m'$$

$$\Rightarrow m' = \frac{T^2K}{4\pi^2} - m_0$$

$$\Rightarrow m' = \frac{0.785\times 19600}{4\times(3.1416)^2} - 400\ gm$$

$$\Rightarrow m' = -10.269\,gm$$

$\therefore$ The spring constant is $K = 19600\,gm\,s^{-2}$ and effective mass $m' = -10.269\,gm$.